

combined_librespeech_rover

scalars_rover_librespeech

key	value
skipped_empty_gt	0
step2_stats.n_models	7
step2_stats.spearman	1
step2_stats.kendall_b	1
step2_stats.spearman_exact_p_onetailed	0.0002
step2_stats.kendall_exact_p_onetailed	0.0002
step2_stats.p_method	exact
step2_stats.n_permutations	5040
step2_stats.min_attainable_p	0.0002
step2_stats.pearson	0.9395
step2_stats.pearson_p	0.0017
step2_stats.mean_wer_compression	0.5073
step3_determinism.identical_outputs	2611
step3_determinism.n	2611
step3_determinism.deterministic	True
step4_resolution_true.reference	true
step4_resolution_true.n_boot	10000
step4_resolution_true.n_samples	2611
step4_resolution_true.min_resolved_gap_wer	0.1414
step4_resolution_true.max_unresolved_gap_wer	0.0325
step4_resolution_true.n_resolved	20
step4_resolution_true.n_pairs	21
step4_resolution_pseudo.reference	pseudo
step4_resolution_pseudo.n_boot	10000
step4_resolution_pseudo.n_samples	2611
step4_resolution_pseudo.min_resolved_gap_wer	0.114
step4_resolution_pseudo.max_unresolved_gap_wer	0.0859
step4_resolution_pseudo.n_resolved	18
step4_resolution_pseudo.n_pairs	21

step0_cross_wer_rover_librespeech

	seamless_m4t	whisper_v3	canary	conformer	parakeet	sb	cohere
seamless_m4t	0	2.84	2.21	2.38	2.14	2.56	2.14
whisper_v3	2.84	0	2.35	2.6	2.22	2.81	2.16

	seamless_m4t	whisper_v3	canary	conformer	parakeet	sb	cohere
canary	2.21	2.35	0	1.9	1.27	2.16	1.54
conformer	2.38	2.6	1.9	0	1.6	1.89	1.6
parakeet	2.14	2.22	1.27	1.6	0	1.88	1.29
sb	2.56	2.81	2.16	1.89	1.88	0	1.74
cohere	2.14	2.16	1.54	1.6	1.29	1.74	0

step0_err_corr_rover_librespeech

	seamless_m4t	whisper_v3	canary	conformer	parakeet	sb	cohere
seamless_m4t	1	0.5964	0.6161	0.6072	0.6198	0.5441	0.54
whisper_v3	0.5964	1	0.6137	0.5985	0.6236	0.5265	0.5666
canary	0.6161	0.6137	1	0.6336	0.7444	0.5804	0.5842
conformer	0.6072	0.5985	0.6336	1	0.6947	0.6515	0.61
parakeet	0.6198	0.6236	0.7444	0.6947	1	0.6234	0.6282
sb	0.5441	0.5265	0.5804	0.6515	0.6234	1	0.5721
cohere	0.54	0.5666	0.5842	0.61	0.6282	0.5721	1

step0_flags_rover_librespeech

pair	cross_wer	err_corr	note
canary parakeet	1.27	0.7444	unusually similar hypotheses - suspect shared lab/data/teacher
canary cohere	1.54	0.5842	unusually similar hypotheses - suspect shared lab/data/teacher
conformer parakeet	1.6	0.6947	unusually similar hypotheses - suspect shared lab/data/teacher
conformer cohere	1.6	0.61	unusually similar hypotheses - suspect shared lab/data/teacher
parakeet cohere	1.29	0.6282	unusually similar hypotheses - suspect shared lab/data/teacher

step1_bias_rover_librespeech

model	role	n_samples	spearman_full	spearman_loo	kendall_full	kendall_loo	delta_spearman	delta_kendall
cohere	voter	2611	0.4111	0.4164	0.3932	0.3967	0.0053	0.0034
parakeet	voter	2611	0.5803	0.6001	0.556	0.5732	0.0197	0.0171
whisper_v3	voter	2611	0.6424	0.6707	0.6133	0.6404	0.0282	0.0271
sb	voter	2611	0.6693	0.6875	0.6395	0.6565	0.0182	0.0169
canary	ranked_only(control)	2611	0.6695	0.6695	0.6397	0.6397	0	0
conformer	ranked_only(control)	2611	0.7215	0.7215	0.6891	0.6891	0	0
seamless_m4t	voter	2611	0.7489	0.7758	0.7148	0.7405	0.0268	0.0257

step2_ranking_rover_librespeech

model	role	n_samples	wer_true	cer_true	wer_pseudo_loo	cer_pseudo_loo	rank_true	rank_pseudo	wer_compression
cohere	voter	2611	1.2245	0.3722	1.1257	0.3998	1	1	0.0988
parakeet	voter	2611	1.6677	0.4946	1.2396	0.4156	2	2	0.428
canary	ranked_only	2611	1.8912	0.6029	1.2848	0.4303	3	3	0.6063
conformer	ranked_only	2611	2.0325	0.5964	1.3536	0.4003	4	4	0.6789
sb	voter	2611	2.2063	0.6697	1.6642	0.5276	5	5	0.5421
seamless_m4t	voter	2611	2.5693	0.9011	1.9443	0.7437	6	6	0.625
whisper_v3	voter	2611	2.6018	1.3488	2.0302	1.192	7	7	0.5715

step3_model_dropout_rover_librespeech

model	times_filtered_out	times_in_pool	filtered_out_rate
seamless_m4t	2114	13055	0.1619
sb	1499	13055	0.1148
whisper_v3	1269	13055	0.0972
parakeet	826	13055	0.0633
cohere	669	13055	0.0512

step3_runs_rover_librespeech

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_rover_full_wtn
full_pool	5	2611	0.4025	0.0033	1
LOO_minus_seamless_m4t	4	2611	0.283	0.0044	1
LOO_minus_whisper_v3	4	2611	0.3095	0.0052	1
LOO_minus_parakeet	4	2611	0.3275	0.005	1
LOO_minus_sb	4	2611	0.3052	0.0041	1
LOO_minus_cohere	4	2611	0.334	0.0051	1

step4_pairs_pseudo_rover_librespeech

pair	adjacent	gap_wer	preserved_frac	resolved_95
parakeet < canary	True	0.0452	0.7742	False
canary < conformer	True	0.0687	0.84	False
seamless_m4t < whisper_v3	True	0.0859	0.6245	False
cohere < parakeet	True	0.114	0.985	True
parakeet < conformer	False	0.114	0.9715	True
cohere < canary	False	0.1592	0.9955	True
cohere < conformer	False	0.2279	0.9998	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
sb < seamless_m4t	True	0.2801	1	True
conformer < sb	True	0.3106	1	True
sb < whisper_v3	False	0.366	0.9631	True
canary < sb	False	0.3794	1	True
parakeet < sb	False	0.4246	1	True
cohere < sb	False	0.5386	1	True
conformer < seamless_m4t	False	0.5908	1	True
canary < seamless_m4t	False	0.6595	1	True
conformer < whisper_v3	False	0.6767	1	True
parakeet < seamless_m4t	False	0.7047	1	True
canary < whisper_v3	False	0.7454	1	True
parakeet < whisper_v3	False	0.7906	1	True
cohere < seamless_m4t	False	0.8187	1	True
cohere < whisper_v3	False	0.9046	1	True

step4_pairs_true_rover_librespeech

pair	adjacent	gap_wer	preserved_frac	resolved_95
seamless_m4t < whisper_v3	True	0.0325	0.5275	False
canary < conformer	True	0.1414	0.9871	True
conformer < sb	True	0.1738	0.9965	True
parakeet < canary	True	0.2235	1	True
canary < sb	False	0.3152	1	True
sb < seamless_m4t	True	0.3629	1	True
parakeet < conformer	False	0.3649	1	True
sb < whisper_v3	False	0.3954	0.9733	True
cohere < parakeet	True	0.4432	1	True
conformer < seamless_m4t	False	0.5368	1	True
parakeet < sb	False	0.5387	1	True
conformer < whisper_v3	False	0.5693	0.9993	True
cohere < canary	False	0.6667	1	True
canary < seamless_m4t	False	0.6781	1	True
canary < whisper_v3	False	0.7106	1	True
cohere < conformer	False	0.808	1	True
parakeet < seamless_m4t	False	0.9016	1	True
parakeet < whisper_v3	False	0.9341	1	True
cohere < sb	False	0.9819	1	True
cohere < seamless_m4t	False	1.3448	1	True
cohere < whisper_v3	False	1.3773	1	True

step5_baselines_rover_librespeech

reference	n_models	spearman	kendall_b	spearman_exact_p_onetailed	kendall_exact_p_onetailed	p_method	n_permutations	min_attainable_p	mean_abs_wer_err
LOO ensemble (proposed)	7	1	1	0.0002	0.0002	exact	5040	0.0002	0.5073
full ensemble (biased)	7	1	1	0.0002	0.0002	exact	5040	0.0002	0.7447
single model: seamless_m4t	6	1	1	0.0014	0.0014	exact	720	0.0014	0.4404
LOO ensemble matched to n-1 (vs seamless_m4t)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.4876
single model: whisper_v3	6	1	1	0.0014	0.0014	exact	720	0.0014	0.5616
LOO ensemble matched to n-1 (vs whisper_v3)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.4965
single model: parakeet	6	0.9429	0.8667	0.0083	0.0083	exact	720	0.0014	0.3735
LOO ensemble matched to n-1 (vs parakeet)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.5205
single model: sb	6	0.9429	0.8667	0.0083	0.0083	exact	720	0.0014	0.2294
LOO ensemble matched to n-1 (vs sb)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.5014
single model: cohere	6	1	1	0.0014	0.0014	exact	720	0.0014	0.4138
LOO ensemble matched to n-1 (vs cohere)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.5753

step6_contribution_rover_librespeech

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
seamless_m4t	-0.0153	-0.7143	False	False	8	8
whisper_v3	0.0134	-0.1786	True	False	4	3
parakeet	0.1242	-0.3571	True	False	6	5
sb	0.0497	-0.9643	True	False	9	8
cohere	0.2292	-0.3929	True	False	5	4

step6_pools_rover_librespeech

pool	k	ensemble_wer_vs_truere	ensemble_cer_vs_truere	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
whisper_v3+parakeet+sb+cohere	4	1.362	0.4516	0.2857	0.278	0.2381	0.281	8	8	1.3448	cohere seamless_m4t
seamless_m4t+whisper_v3+parakeet+sb+cohere	5	1.3773	0.4195	1	0.0002	1	0.0002	0	0	0	
seamless_m4t+parakeet+sb+cohere	4	1.3907	0.4368	0.8214	0.0171	0.619	0.0345	4	3	0.3649	parakeet conformer
parakeet+sb+cohere	3	1.4193	0.5426								
seamless_m4t+whisper_v3+parakeet+cohere	4	1.427	0.4614	0.0357	0.4817	0.1429	0.3863	9	8	1.3773	cohere whisper_v3
seamless_m4t+whisper_v3+sb+cohere	4	1.5015	0.5036	0.6429	0.0694	0.4286	0.1194	6	5	0.808	cohere conformer
seamless_m4t+parakeet+cohere	3	1.5416	0.6534								
seamless_m4t+whisper_v3+parakeet+sb	4	1.6065	0.5303	0.6071	0.0833	0.5238	0.0681	5	4	0.9341	parakeet whisper_v3
seamless_m4t+sb+cohere	3	1.6657	0.6617								

pool	k	ensemble_wer_vs_t rue	ensemble_cer_vs_tr ue	ranking_spear man	ranking_exact _p	ranking_kendall _b	ranking_kendall_exact_ p	n_pairs_wro ng	n_pairs_wrong_resolvab le	worst_wrong_gap_ wer	worst_wrong_pair
seamless_m4t+parakeet+sb	3	1.7842	0.734								
whisper_v3+parakeet+cohere	3	2.2274	1.3181								
whisper_v3+sb+cohere	3	2.3076	1.3668								
whisper_v3+parakeet+sb	3	2.426	1.4073								
seamless_m4t+whisper_v3+cohere	3	2.5196	1.4874								
seamless_m4t+whisper_v3+parakeet	3	2.6457	1.5246								
seamless_m4t+whisper_v3+sb	3	2.7489	1.5603								

step6_wrong_pairs_rover_librespeech

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
seamless_m4t+whisper_v3+parakeet+cohere	4	cohere whisper_v3	cohere	whisper_v3	1.2245	2.6018	2.1936	2.17	1.3773	0.0235	True
whisper_v3+parakeet+sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	1.2245	2.5693	1.9605	1.9443	1.3448	0.0161	True
seamless_m4t+whisper_v3+parakeet+cohere	4	cohere sb	cohere	sb	1.2245	2.2063	2.1936	1.6642	0.9819	0.5294	True
seamless_m4t+whisper_v3+parakeet+sb	4	parakeet whisper_v3	parakeet	whisper_v3	1.6677	2.6018	2.3396	2.2856	0.9341	0.054	True
seamless_m4t+whisper_v3+parakeet+cohere	4	parakeet whisper_v3	parakeet	whisper_v3	1.6677	2.6018	2.261	2.17	0.9341	0.091	True
whisper_v3+parakeet+sb+cohere	4	parakeet seamless_m4t	parakeet	seamless_m4t	1.6677	2.5693	2.0963	1.9443	0.9016	0.152	True
seamless_m4t+whisper_v3+parakeet+cohere	4	cohere conformer	cohere	conformer	1.2245	2.0325	2.1936	1.4428	0.808	0.7507	True
seamless_m4t+whisper_v3+sb+cohere	4	cohere conformer	cohere	conformer	1.2245	2.0325	2.2396	1.4742	0.808	0.7654	True
whisper_v3+parakeet+sb+cohere	4	cohere conformer	cohere	conformer	1.2245	2.0325	1.9605	1.37	0.808	0.5905	True
seamless_m4t+whisper_v3+parakeet+cohere	4	cohere canary	cohere	canary	1.2245	1.8912	2.1936	1.2844	0.6667	0.9091	True
seamless_m4t+whisper_v3+sb+cohere	4	cohere canary	cohere	canary	1.2245	1.8912	2.2396	1.478	0.6667	0.7616	True
whisper_v3+parakeet+sb+cohere	4	cohere canary	cohere	canary	1.2245	1.8912	1.9605	1.3509	0.6667	0.6095	True
seamless_m4t+whisper_v3+parakeet+cohere	4	parakeet sb	parakeet	sb	1.6677	2.2063	2.261	1.6642	0.5387	0.5968	True
seamless_m4t+whisper_v3+sb+cohere	4	cohere parakeet	cohere	parakeet	1.2245	1.6677	2.2396	1.2396	0.4432	1	True
seamless_m4t+whisper_v3+parakeet+sb	4	sb whisper_v3	sb	whisper_v3	2.2063	2.6018	2.7561	2.2856	0.3954	0.4705	True
seamless_m4t+whisper_v3+sb+cohere	4	sb whisper_v3	sb	whisper_v3	2.2063	2.6018	2.7196	2.2989	0.3954	0.4207	True
whisper_v3+parakeet+sb+cohere	4	sb whisper_v3	sb	whisper_v3	2.2063	2.6018	2.5282	2.1451	0.3954	0.3831	True
seamless_m4t+whisper_v3+parakeet+sb	4	parakeet conformer	parakeet	conformer	1.6677	2.0325	2.3396	1.4233	0.3649	0.9163	True
seamless_m4t+whisper_v3+parakeet+cohere	4	parakeet conformer	parakeet	conformer	1.6677	2.0325	2.261	1.4428	0.3649	0.8182	True
seamless_m4t+parakeet+sb+cohere	4	parakeet conformer	parakeet	conformer	1.6677	2.0325	1.4349	1.3376	0.3649	0.0973	True
whisper_v3+parakeet+sb+cohere	4	parakeet conformer	parakeet	conformer	1.6677	2.0325	2.0963	1.37	0.3649	0.7263	True
whisper_v3+parakeet+sb+cohere	4	sb seamless_m4t	sb	seamless_m4t	2.2063	2.5693	2.5282	1.9443	0.3629	0.5839	True
seamless_m4t+whisper_v3+parakeet+sb	4	parakeet canary	parakeet	canary	1.6677	1.8912	2.3396	1.3527	0.2235	0.9869	True
seamless_m4t+whisper_v3+parakeet+cohere	4	parakeet canary	parakeet	canary	1.6677	1.8912	2.261	1.2844	0.2235	0.9766	True
seamless_m4t+parakeet+sb+cohere	4	parakeet canary	parakeet	canary	1.6677	1.8912	1.4349	1.3586	0.2235	0.0763	True
whisper_v3+parakeet+sb+cohere	4	parakeet canary	parakeet	canary	1.6677	1.8912	2.0963	1.3509	0.2235	0.7454	True
seamless_m4t+whisper_v3+sb+cohere	4	canary conformer	canary	conformer	1.8912	2.0325	1.478	1.4742	0.1414	0.0038	True
seamless_m4t+parakeet+sb+cohere	4	canary conformer	canary	conformer	1.8912	2.0325	1.3586	1.3376	0.1414	0.021	True
seamless_m4t+whisper_v3+parakeet+sb	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	2.5693	2.6018	2.8387	2.2856	0.0325	0.5531	False
seamless_m4t+whisper_v3+parakeet+cohere	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	2.5693	2.6018	2.8249	2.17	0.0325	0.6548	False
seamless_m4t+whisper_v3+sb+cohere	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	2.5693	2.6018	2.8517	2.2989	0.0325	0.5528	False

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
seamless_m4t+parakeet+sb+cohere	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	2.5693	2.6018	2.0632	2.0302	0.0325	0.0329	False

step7_floor_rover_librespeech

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
cohere	0.8407	0.3967	0.2127	416	0.409
parakeet	0.792	0.5732	0.3168	543	0.5921
canary	0.7641	0.6397	0.3308	616	0.6584
conformer	0.7472	0.6891	0.3543	660	0.7066
whisper_v3	0.7465	0.6404	0.3512	662	0.6653
sb	0.7442	0.6565	0.4059	668	0.6802
seamless_m4t	0.6967	0.7405	0.4865	792	0.7569